8086 Machine Codes

Dept. of Computer Science and Engineering

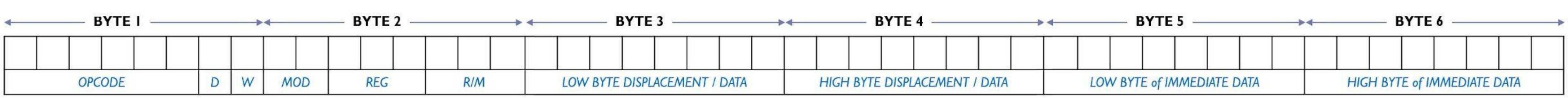
Instruction template

- For 8085: Just look up the hex code for each instruction.
- For 8086 it is not simple.
- E.g 32 ways to specify the source in **MOV CX, source**.
- MOV CX, **source**

a 16-bit register (8 in number)

a memory location (24 possible memory addressing modes)

- Each of these 32 instructions require different binary code.
- Impractical to list them all in a table.
- Instruction templates help code the instruction properly.



Instruction template (6 bytes)

BYTE 1								BYTE 2							
OPCODE						*D*	*W*	*MOD*		*REG*			*R/M*		

BYTE 3								BYTE 4							
LOW BYTE DISPLACEMENT / DATA								*HIGH BYTE DISPLACEMENT / DATA*							

BYTE 5								BYTE 6							
LOW BYTE of IMMEDIATE DATA								*HIGH BYTE of IMMEDIATE DATA*							

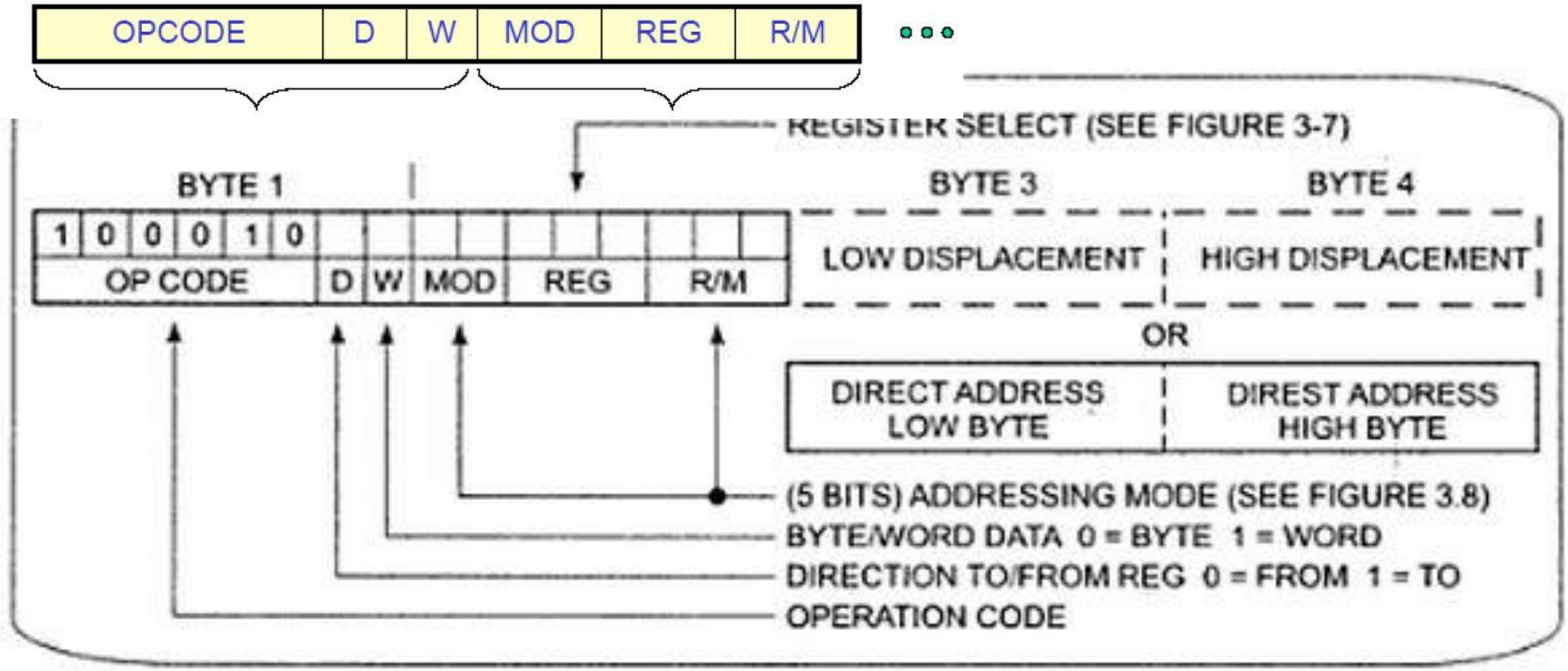
An instruction after conversion can have 1 to 6 bytes long of machine code

Constructing Machine Codes for 8086

- Each instruction in 8086 is associated with the binary code.
- You need to locate the codes appropriately.
- Most of the time this work will be done by assembler
- The things needed to keep in mind is:
  - Instruction templates and coding formats
  - MOD and R/M Bit patterns for particular instruction

MOV Instruction Coding

- MOV data from a register to a register/to a memory location or from a memory location to a register. (**Operation Code of MOV: 100010**)



OPCODE
D
W
MOD
REG
R/M
REGISTER SELECT (SEE FIGURE 3-7)
BYTE 1
BYTE 3
BYTE 4
1 0 0 0 1 0
OP CODE
D
W
MOD
REG
R/M
LOW DISPLACEMENT
HIGH DISPLACEMENT
OR
DIRECT ADDRESS LOW BYTE
DIREST ADDRESS HIGH BYTE
(5 BITS) ADDRESSING MODE (SEE FIGURE 3.8)
BYTE/WORD DATA 0 = BYTE 1 = WORD
DIRECTION TO/FROM REG 0 = FROM 1 = TO
OPERATION CODE

MOD and R/M Field

- 2-bit Mode (MOD) and 3-bit Register/Memory (R/M) fields specify the other operand.
- Also specify the addressing mode.

MOD and R/M Field

RM \ MOD	00	01	10	11	
				W = 0	W = 1
000	[BX] + [SI]	[BX] + [SI] + d8	[BX] + [SI] + d16	AL	AX
001	[BXI + [DI]	[BX] + [DI] + d8	[BX] + [DI] + d16	CL	CX
010	[BP] + [SI]	[BP] + [SI] + d8	[BP] + [SI] + d16	DL	DX
011	[BP] + [DI]	[BP] + [DI] + d8	[BP] + [DI] + d16	BL	BX
100	[SI]	[SI] + d8	[SI] +d16	AH	SP
101	[DI]	[DI] + d8	[DI] +d16	CH	BP
110	d16 (direct address)	[BP] + d8	[BP] +d16	DH	SI
111	[BX]	[BX] + d8	[BX] + d16	BH	DI

MOD and R/M Field

- If the other operand in the instruction is also one of the eight register then put in 11 for MOD bits in the instruction code.

- If the other operand is memory location, there are 24 ways of specifying how the execution unit should compute the effective address of the operand in the main memory.

- If the effective address specified in the instruction contains displacement less than 256 along with the reference to the contents of the register then put in 01 as the MOD bits.

- If the expression for the effective address contains a displacement which is too large to fit in 8 bits then out in 10 in MOD bits.

REG Field

- REG field is used to identify the register of the one operand

REG	W = 0	W = 1
000	AL	AX
001	CL	CX
010	DL	DX
011	BL	BX
100	AH	SP
101	CH	BP
110	DH	SI
111	BH	DI

Instruction template

BYTE 1			BYTE 2		
OPCODE	*D*	*W*	*MOD*	*REG*	*R/M*

6 bits of MOV, ADD etc

D - direction
If **D=0**, then direction is from a register (source) If **D=1**, then direction is to a register (destination)

W - word
If **W=0**, then only a byte is being transferred (8 bits)
If **W=1**, them a whole word is being transferred (16 bits)

MODE	OPERAND NATURE	
00	Memory with no displacement	*MOV AX, [BX]*
01	Memory with 8-bit displacement	*MOV AX, [BX + 12h]*
10	Memory with 16-bit displacement	*MOV AX, [BX + 1234h]*
11	Both are registers	*MOV AX, BX*

- 34h here is an 8-bit displacement
- [BX+34h] is a memory/offset address

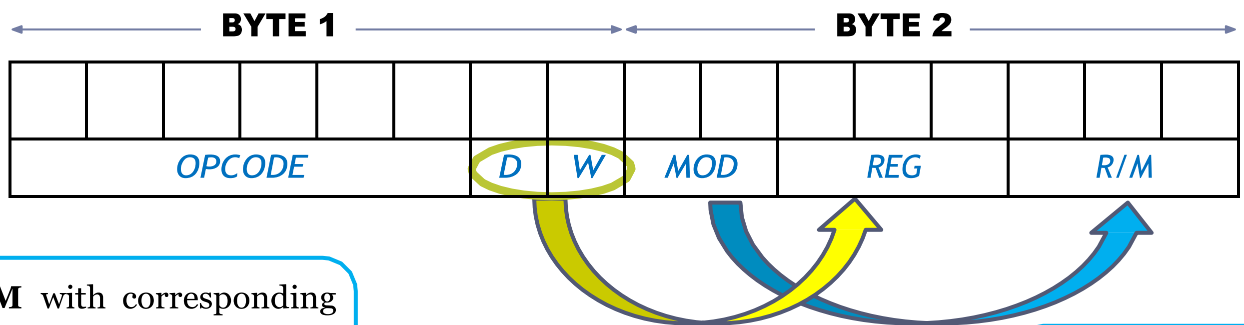
MOV [BX + $\boxed{34h}$], AL

Instruction template

MOV AX, 1234 h

- 1234h here is a 16-bit immediate data value

Instruction template



- Value for **R/M** with corresponding MOD value
- Value for REG with corresponding **W** value and the register considered in D

Check column that matches with MOD value

RM \ MOD	00	01	10	11 W = 0	11 W = 1
000	[BX] + [SI]	[BX] + [SI] + d8	[BX] + [SI] + d16	AL	AX
001	[BX] + [DI]	[BX] + [DI] + d8	[BX] + [DI] + d16	CL	CX
010	[BP] + [SI]	[BP] + [SI] + d8	[BP] + [SI] + d16	DL	DX
011	[BP] + [DI]	[BP] + [DI] + d8	[BP] + [DI] + d16	BL	BX
100	[SI]	[SI] + d8	[SI] +d16	AH	SP
101	[DI]	[DI] + d8	[DI] +d16	CH	BP
110	d16 (direct address)	[BP] + d8	[BP] +d16	DH	SI
111	[BX]	[BX] + d8	[BX] + d16	BH	DI

Example 1

- **MOV 8B43H [SI], DH:** Copy a byte from DH to memory with 16 bit displacement given the opcode for MOV=100010

BYTE 1								BYTE 2							
1	0	0	0	1	0	0	0	1	0						
OPCODE						D	W	MOD		REG			R/M		

Therefore D=0

source -- 8 bits (not a word size)

Therefore W=0

MOV [SI + 8B43H], DH

MODE	OPERAND NATURE
00	Memory with no displacement
01	Memory with 8-bit displacement
10	Memory with 16-bit displacement
11	Both are registers

BYTE 3	BYTE 4

Example 1

LOW BYTE DISPLACEMENT / DATA								*HIGH BYTE DISPLACEMENT / DATA*							

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- **MOV 8B43H [SI], DH:** Copy a byte from DH to memory with 16 bit displacement given the opcode for MOV=100010

BYTE 1								BYTE 2							
1	0	0	0	1	0	0	0	1	0	1	1	0			
OPCODE						*D*	*W*	*MOD*		*REG*			*R/M*		

MOV [SI + 8B43H] ,
DH

RM \ MOD	00	01	10	11	
				W = 0	W = 1
000	[BX] + [SI]	[BX] + [SI] + d8	[BX] + [SI] + d16	AL	AX
001	[BX] + [DI]	[BX] + [DI] + d8	[BX] + [DI] + d16	CL	CX
010	[BP] + [SI]	[BP] + [SI] + d8	[BP] + [SI] + d16	DL	DX
011	[BP] + [DI]	[BP] + [DI] + d8	[BP] + [DI] + d16	BL	BX
100	[SI]	[SI] + d8	[SI] + d16	AH	SP
101	[DI]	[DI] + d8	[DI] +d16	CH	BP
110	d16 (direct address)	[BP] + d8	[BP] +d16	DH	SI
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BYTE 1								BYTE 2							
1	0	0	0	1	0	0	0	1	0	1	1	0	1	0	0
OPCODE						D	W	MOD		REG			R/M		

MOV [SI + 8B43H] , DH

MOD / RM	00	01	10	11 W = 0	11 W = 1
000	[BX] + [SI]	[BX] + [SI] + d8	[BX] + [SI] + d16	AL	AX
001	[BX] + [DI]	[BX] + [DI] + d8	[BX] + [DI] + d16	CL	CX
010	[BP] + [SI]	[BP] + [SI] + d8	[BP] + [SI] + d16	DL	DX
011	[BP] + [DI]	[BP] + [DI] + d8	[BP] + [DI] + d16	BL	BX
100	[SI]	[SI] + d8	[SI] +d16	AH	SP
101	[DI]	[DI] + d8	[DI] +d16	CH	BP
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111	[BX]	[BX] + d8	[BX] + d16	BH	DI

Example 1

- **MOV 8B43H [SI], DH:** Copy a byte from DH to memory with 16 bit displacement given the opcode for MOV=100010

BYTE 1								BYTE 2							
1	0	0	0	1	0	0	0	1	0	1	1	0	1	0	0
OPCODE						*D*	*W*	*MOD*		*REG*			*R/M*		

MOV [SI + 8B43H] , DH

BYTE 3								BYTE 4							
0	1	0	0	0	0	1	1	1	0	0	0	1	0	1	1
LOW BYTE DISPLACEMENT								*HIGH BYTE DISPLACEMENT*							

Machine Code: *1000 1000 1011 0100 0100 0011 1000 1011*$_2$ *or 88 B4 43 8B*$_{16}$

Example 2

- **MOV AX, BX:** given the opcode for MOV=100010

BYTE 1								BYTE 2							
1	0	0	0	1	0	1	1	1	1						
OPCODE						*D*	*W*	*MOD*		*REG*			*R/M*		

AX considered which is a destination operand, therefore D = 0

destination -- 16 bits (a word size)

MOV AX, BX

AX is 16-bit long, therefore W = 1

MODE	OPERAND NATURE
00	Memory with no displacement
01	Memory with 8-bit displacement
10	Memory with 16-bit displacement
11	Both are registers

Example 2

- **MOV AX, BX:** given the opcode for MOV=100010

BYTE 1								BYTE 2							
1	0	0	0	1	0	1	1	1	1	0	0	0			
OPCODE						*D*	*W*	*MOD*		*REG*			*R/M*		

RM \ MOD	00	01	10	11	
				W = 0	W = 1
000	[BX] + [SI]	[BX] + [SI] + d8	[BX] + [SI] + d16	AL	AX
001	[BXI + [DI]	[BX] + [DI] + d8	[BX] + [DI] + d16	CL	CX
010	[BP] + [SI]	[BP] + [SI] + d8	[BP] + [SI] + d16	DL	DX
011	[BP] + [DI]	[BP] + [DI] + d8	[BP] + [DI] + d16	BL	BX
100	[SI]	[SI] + d8	[SI] +d16	AH	SP
101	[DI]	[DI] + d8	[DI] +d16	CH	BP
110	d16 (direct address)	[BP] + d8	[BP] +d16	DH	SI
111	[BX]	[BX] + d8	[BX] + d16	BH	DI

Example 2

Machine Code: *1000 1011 1100 0011*$_2$ *or* *8B C3*$_{16}$

- **MOV AX, BX:** given the opcode for MOV=100010

BYTE 1 BYTE 2

1	0	0	0	1	0	1	1	1	1	0	0	0	0	1	1
OPCODE						*D*	*W*	*MOD*		*REG*			*R/M*		

RM \ MOD	00	01	10	11	
				W = 0	W = 1
000	[BX] + [SI]	[BX] + [SI] + d8	[BX] + [SI] + d16	AL	AX
001	[BX] + [DI]	[BX] + [DI] + d8	[BX] + [DI] + d16	CL	CX
010	[BP] + [SI]	[BP] + [SI] + d8	[BP] + [SI] + d16	DL	DX
011	[BP] + [DI]	[BP] + [DI] + d8	[BP] + [DI] + d16	BL	BX
100	[SI]	[SI] + d8	[SI] +d16	AH	SP
101	[DI]	[DI] + d8	[DI] +d16	CH	BP
110	d16 (direct address)	[BP] + d8	[BP] +d16	DH	SI
111	[BX]	[BX] + d8	[BX] + d16	BH	DI

QUIZ

Compute the machine code for the following using the table below

and the opcode for MOV as100010

a) MOV AX, 5E9Ch

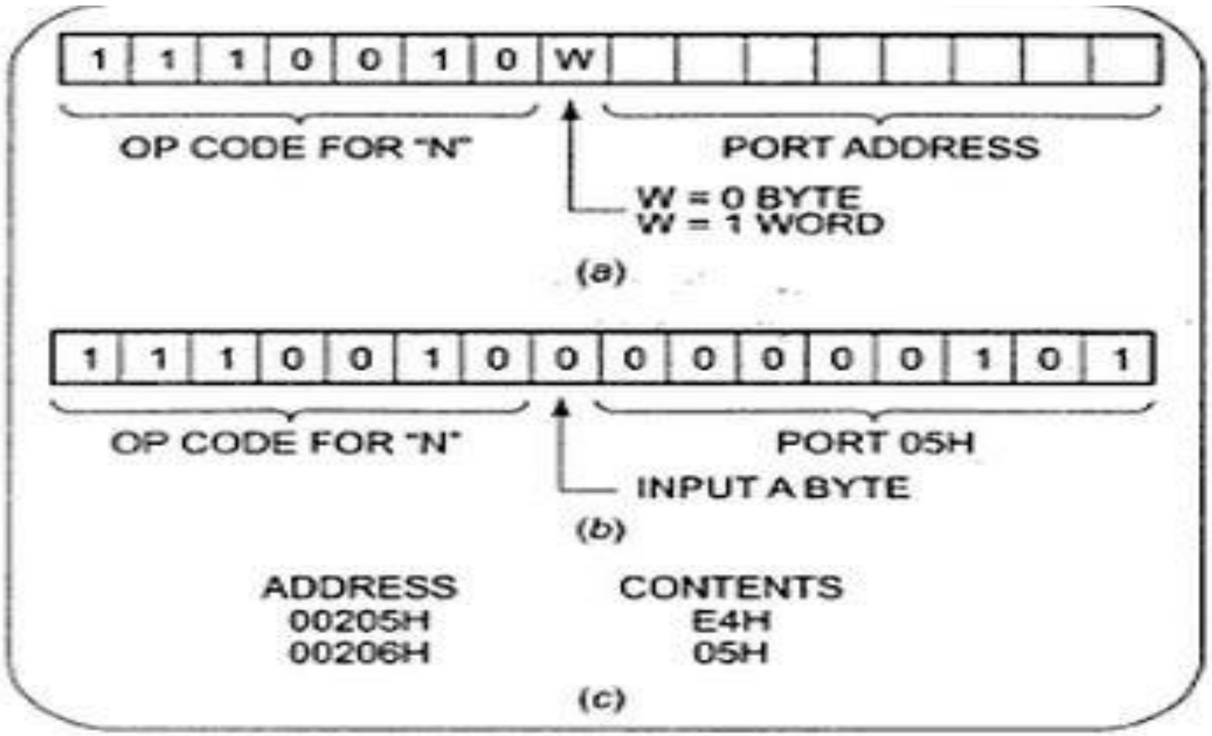
b) MOV DH, [BP+SI+7Dh]

RM \ MOD	00	01	10	11	
				W = 0	W = 1
000	[BX] + [SI]	[BX] + [SI] + d8	[BX] + [SI] + d16	AL	AX
001	[BX] + [DI]	[BX] + [DI] + d8	[BX] + [DI] + d16	CL	CX
010	[BP] + [SI]	[BP] + [SI] + d8	[BP] + [SI] + d16	DL	DX
011	[BP] + [DI]	[BP] + [DI] + d8	[BP] + [DI] + d16	BL	BX
100	[SI]	[SI] + d8	[SI] +d16	AH	SP
101	[DI]	[DI] + d8	[DI] +d16	CH	BP
110	d16 (direct address)	[BP] + d8	[BP] +d16	DH	SI
111	[BX]	[BX] + d8	[BX] + d16	BH	DI

Instruction Template

- The Intel literature shows two different formats for coding 8086 instructions.
- Instruction templates helps you to code the instruction properly.
- **Example:**

IN AL, 05H



Example

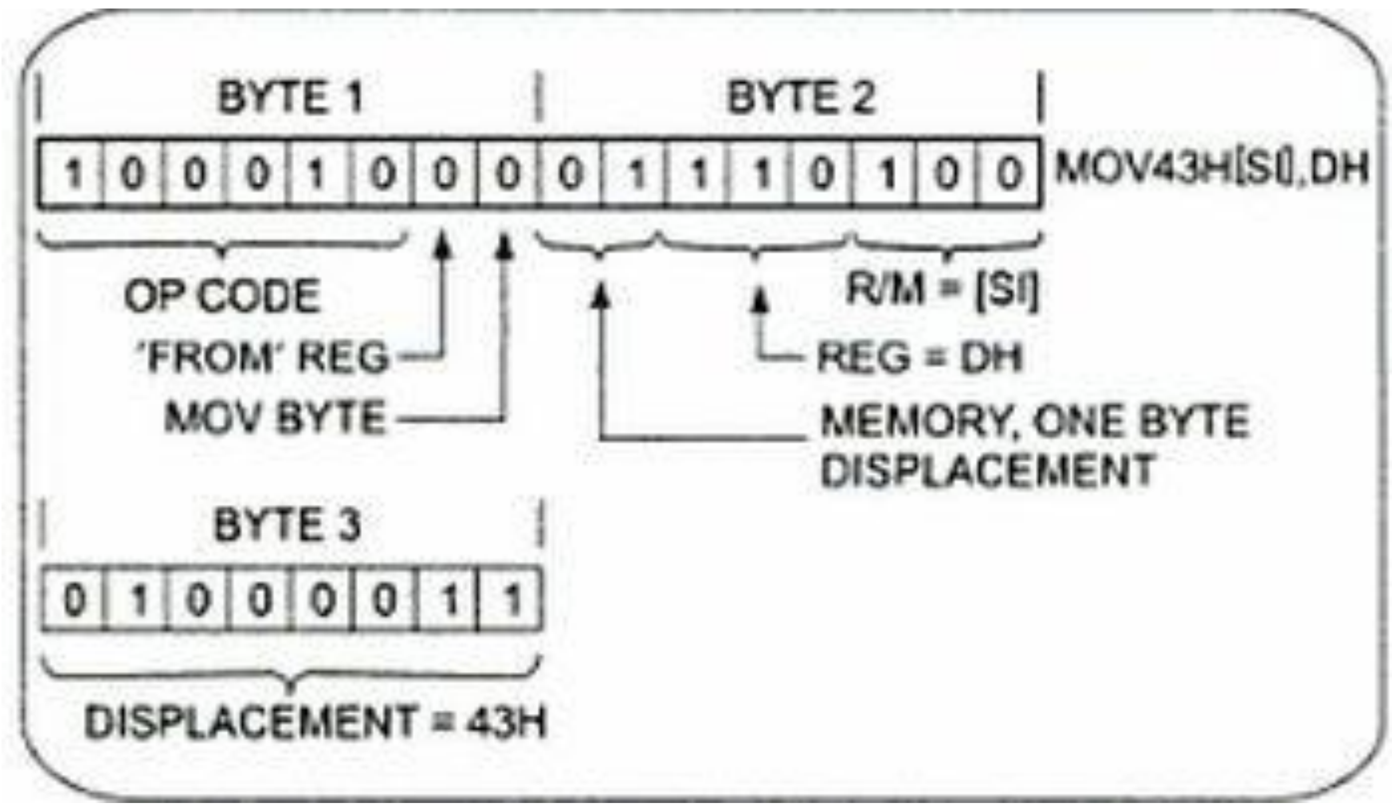
- MOV BL,AL
- Opcode for MOV = 100010
- We'll encode AL so
  - D = 0 (AL source operand)
- W bit = 0 (8-bits)
- MOD = 11 (register mode)
- REG = 000 (code for AL)
- R/M = 011

OPCODE	D	W	MOD	REG	R/M
100010	0	0	11	000	011

Example

- **MOV 43H [SI], DH:** Copy a byte from DH register to memory location.

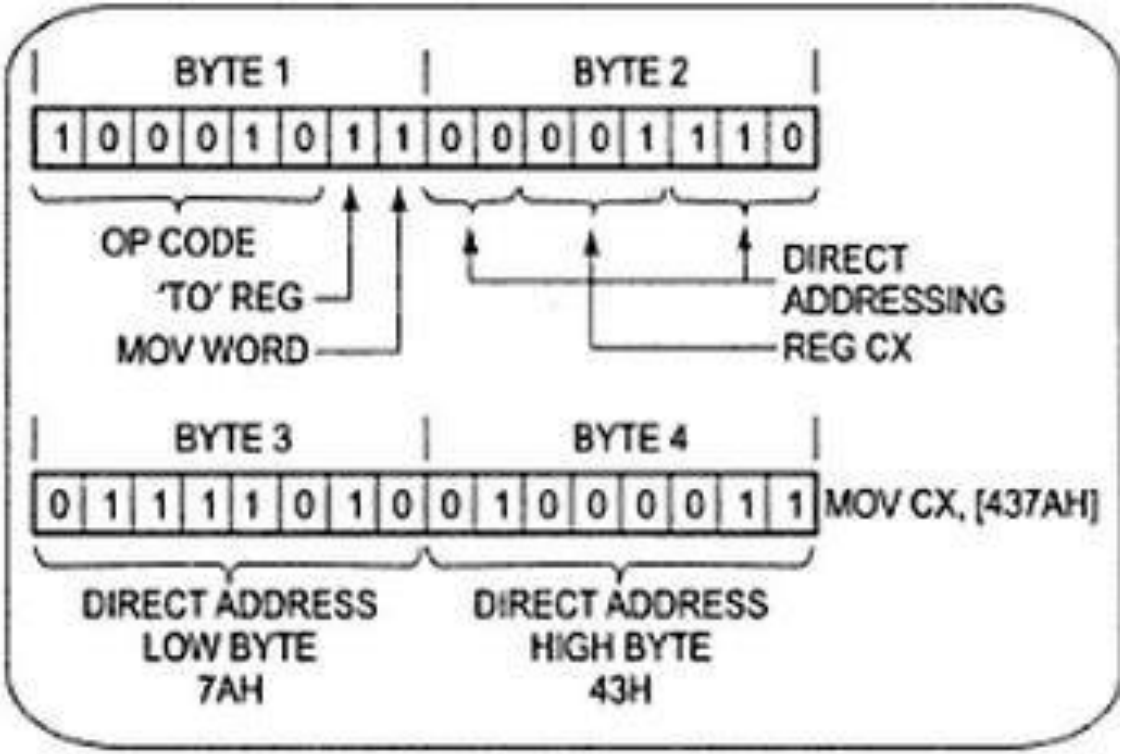
Example



BYTE 1
BYTE 2
1 0 0 0 1 0 0 0 0 1 1 1 0 1 0 0
MOV43H[SI],DH
OP CODE
'FROM' REG
MOV BYTE
R/M = [SI]
REG = DH
MEMORY, ONE BYTE
DISPLACEMENT
BYTE 3
0 1 0 0 0 0 1 1
DISPLACEMENT = 43H

Example 3

- **MOV CX, [437AH]:** Copy the contents of the two memory locations to the register CX.


BYTE 1
BYTE 2
1 0 0 0 1 0 1 1 0 0 0 0 1 1 1 0
OP CODE
'TO' REG
MOV WORD
DIRECT
ADDRESSING
REG CX
BYTE 3
BYTE 4
0 1 1 1 1 0 1 0 0 1 0 0 0 0 1 1
MOV CX, [437AH]
DIRECT ADDRESS
LOW BYTE
7AH
DIRECT ADDRESS
HIGH BYTE
43H

QUIZ

Compute the machine code for the following using the table below

and the opcode for MOV as100010

a) MOV AX, 5E9Ch

b) MOV DH, [BP+SI+7Dh]

MOD / RM	00	01	10	11	
				W = 0	W = 1
000	[BX] + [SI]	[BX] + [SI] + d8	[BX] + [SI] + d16	AL	AX
001	[BXI + [DI]	[BX] + [DI] + d8	[BX] + [DI] + d16	CL	CX
010	[BP] + [SI]	[BP] + [SI] + d8	[BP] + [SI] + d16	DL	DX
011	[BP] + [DI]	[BP] + [DI] + d8	[BP] + [DI] + d16	BL	BX
100	[SI]	[SI] + d8	[SI] +d16	AH	SP
101	[DI]	[DI] + d8	[DI] +d16	CH	BP
110	d16 (direct address)	[BP] + d8	[BP] +d16	DH	SI
111	[BX]	[BX] + d8	[BX] + d16	BH	DI

Thank You !!!